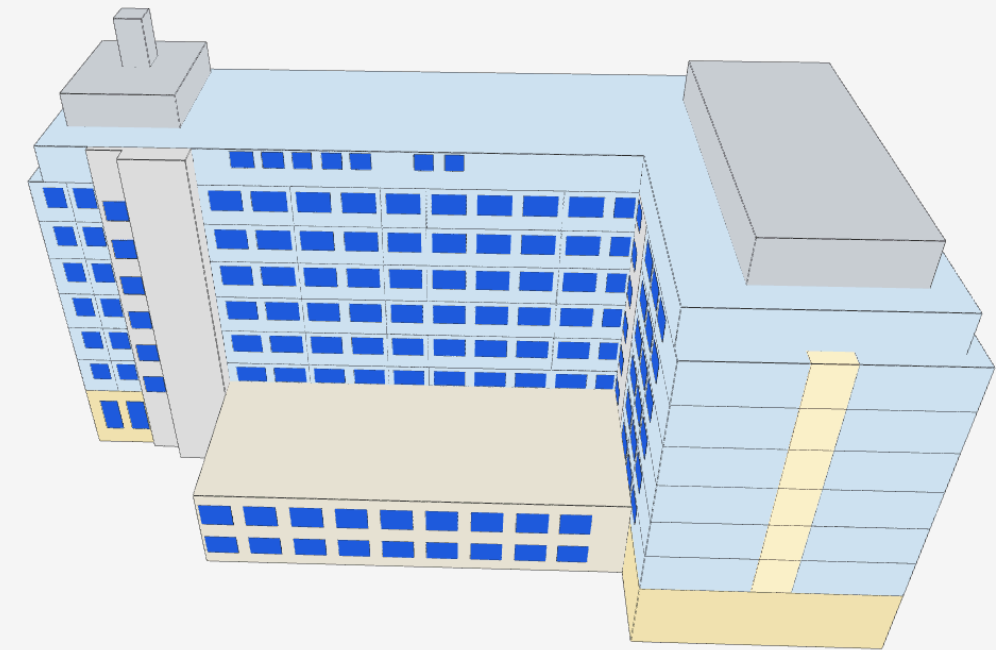


面向设计前期的 仿真轻量 BIM Agent

A lightweight BIM agent
for early-stage design simulation



多模态输入 → 轻量 BIM → 多物理场仿真

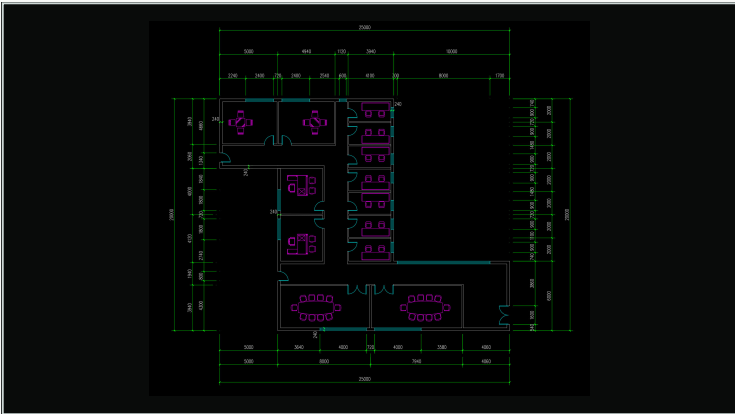
Multimodal inputs · Lightweight BIM · Multiphysics simulation

VOIMATALO

拖动旋转 / Drag to rotate

让建筑仿真跟上方案迭代

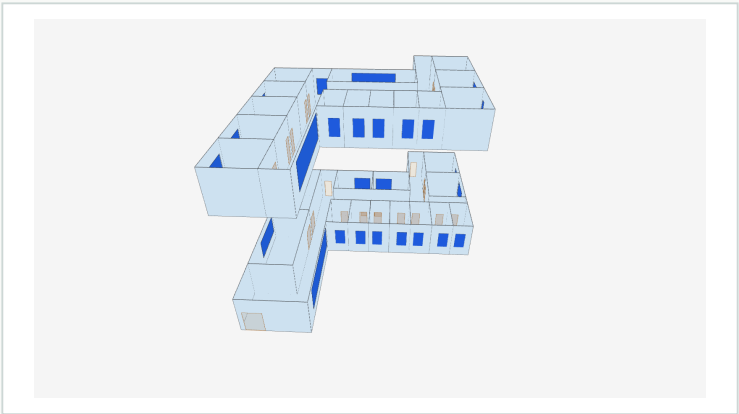
Bringing building simulation into the design iteration loop



方案与图纸

Design & drawings

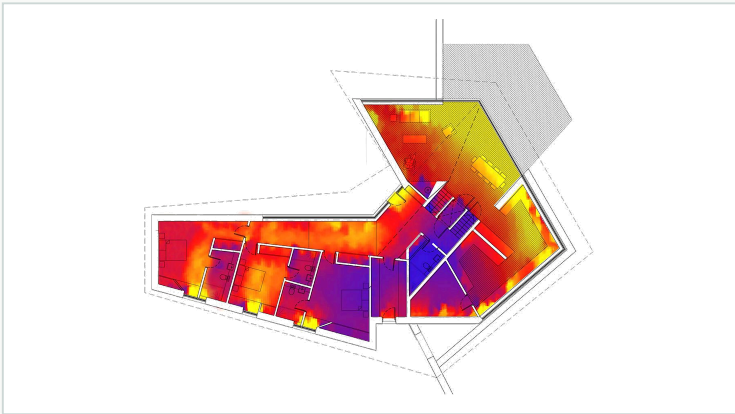
人工识读
Interpret



几何模型

Geometry model

专业配置
Configure



仿真与分析

Simulation & analysis

几何整理
Geometry cleanup



分区与物性
Zoning & properties



后端模型配置
Backend setup



计算与结果回传
Run & return results

长链路 · 依赖人工 · 反馈滞后

Long workflow · Manual effort · Delayed feedback

研究目标
Our aim

建立轻量的公共仿真底座

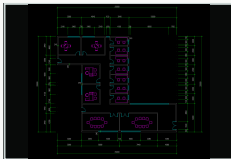
A shared lightweight foundation for building simulation

设计 ⇄ 轻量 BIM ⇄ 仿真

Design · Lightweight BIM · Simulation

多模态输入，轻量 BIM 输出

Multimodal inputs to lightweight BIM



图纸 / Drawings



图像 / Images



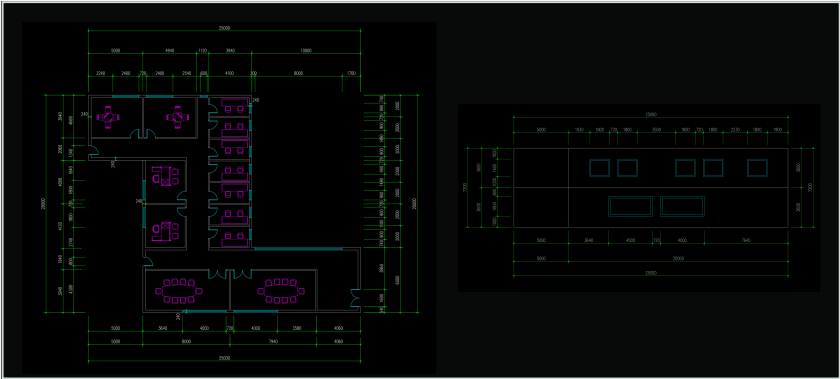
三维体量 / 3D meshes



文字意图 / Design intent



BIM Agent 多模态建模流程 / Multimodal modelling pipeline



还原建模
Reconstruction



部分推理建模
Partial inference



完全推理建模
Full inference

↓ 轻量 BIM Lightweight BIM

Aerial image: Harria / Wikimedia Commons · CC BY-SA 4.0

从信息理解，到公共仿真底座

From input understanding to a shared simulation foundation



↑

根据模型与工具反馈迭代

Iterate with model & tool feedback

模型能力

Model capabilities

几何工具

Geometry tools

交互补充

User input

目标与输入

Goals & inputs

轻量 BIM

Lightweight BIM

仿真插件

Simulation plugins

EnergyPlus

热与能耗 / Thermal & energy

● 已接通 / Connected

扩展接口 / Extensible interfaces

Radiance

采光 / Daylighting

OpenFOAM

气流 / Airflow

Pachyderm

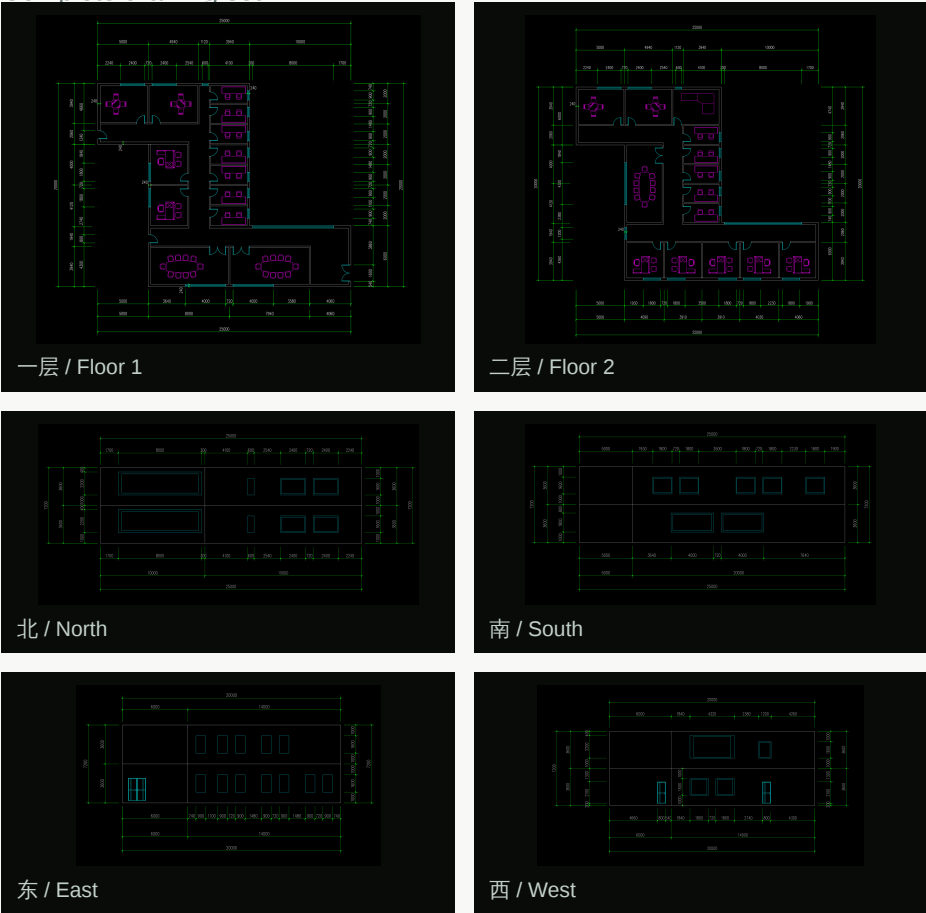
声学 / Acoustics

完整建筑图纸 → 轻量 BIM

A complete architectural drawing set to lightweight BIM

完整图纸输入

Complete drawing set

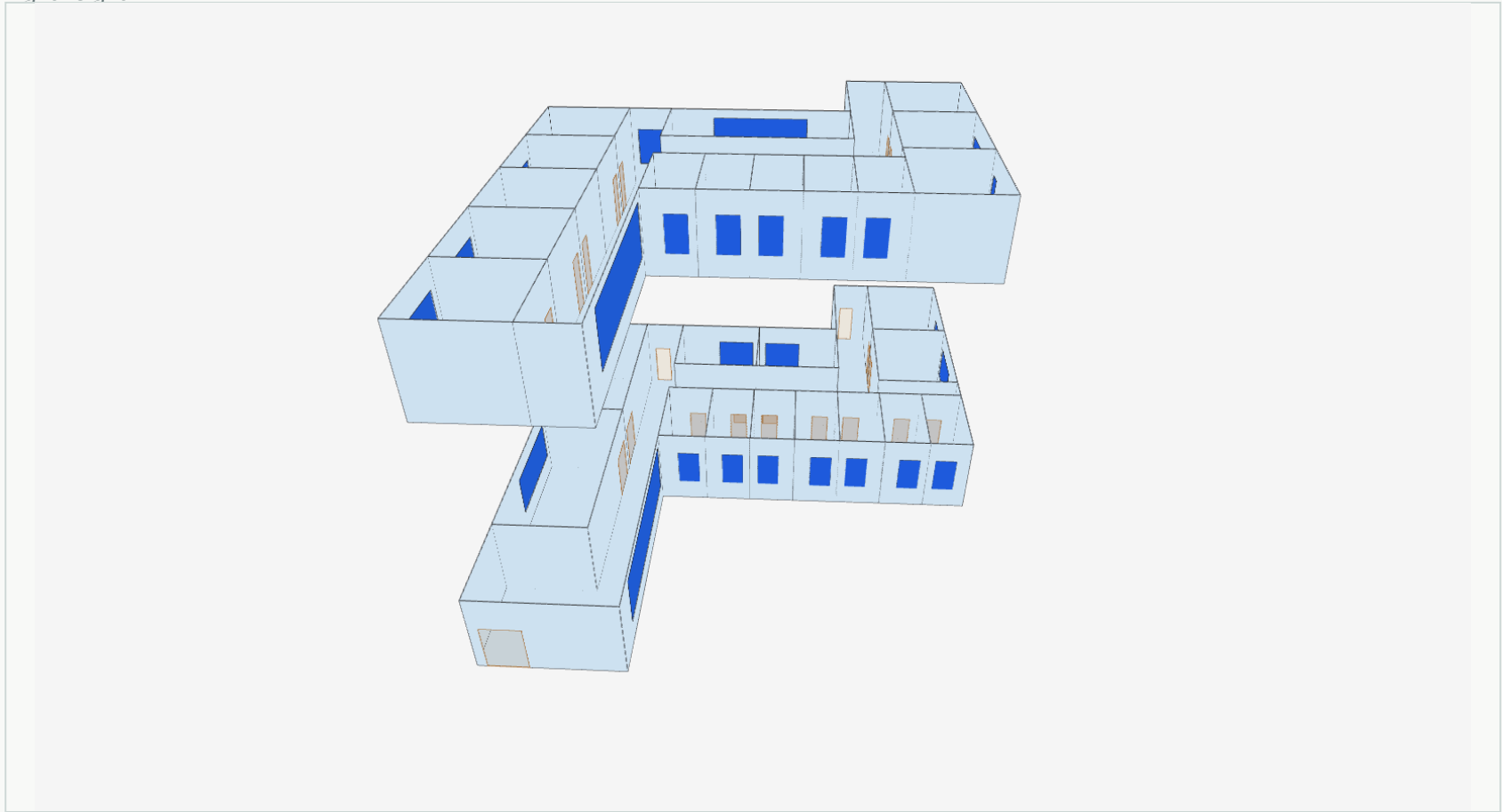


2 张平面 + 4 张立面

2 floor plans + 4 elevations

轻量 BIM

Lightweight BIM



29 空间 Spaces

31 窗 Windows

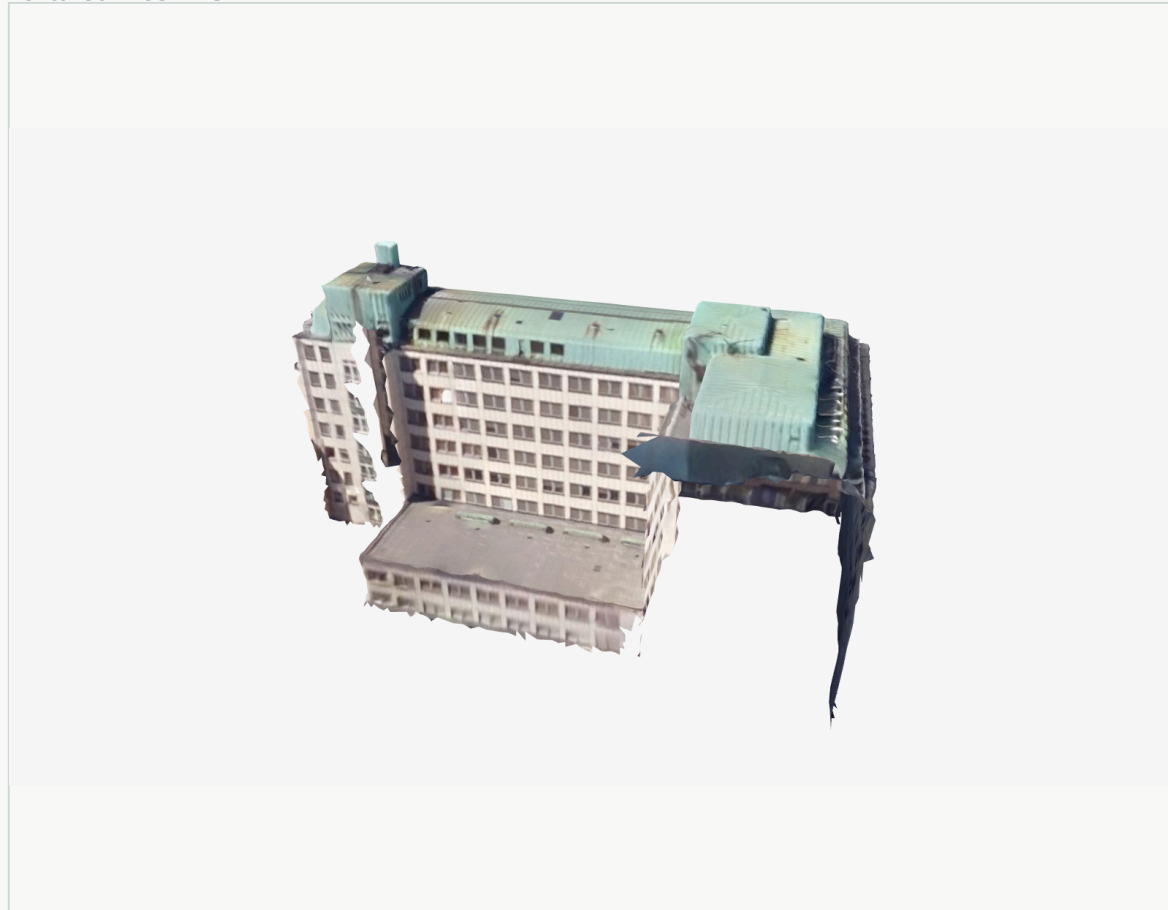
30 门 Doors

真实贴图体量 → 轻量 BIM

A real textured mesh to lightweight BIM

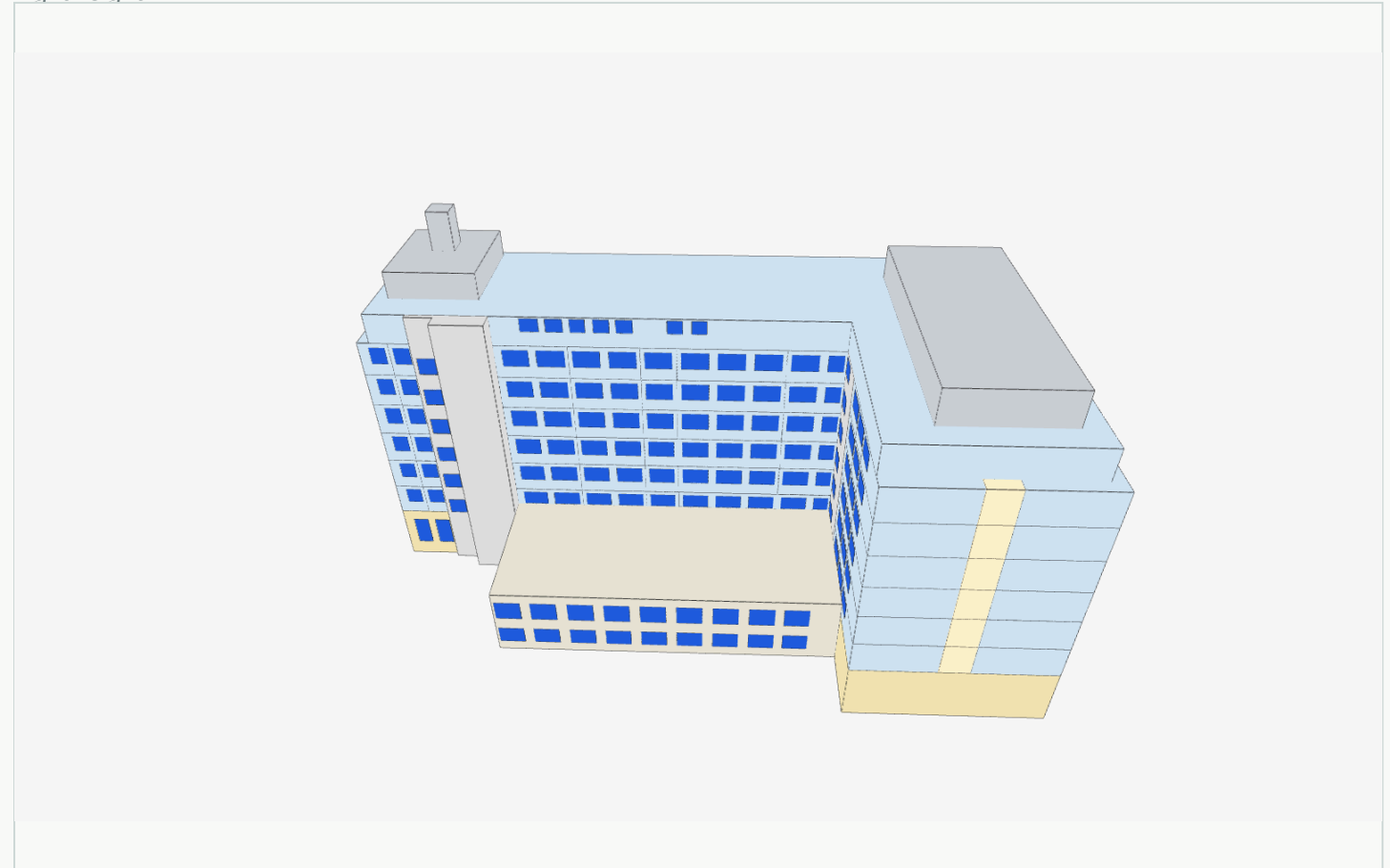
真实贴图体量

Textured mesh · GLB



轻量 BIM

Lightweight BIM



Voimatalo · Helsinki

摄影测量 / Photogrammetry · SUM / Helsinki

8 主楼层数 Main storeys

322 窗组 Window groups

01

提升空间
理解能力

Stronger spatial
understanding

空间 · 连接 · 建筑逻辑

Spaces · Connectivity
Architectural logic

02

提高复杂建筑
稳定性

Robustness on
complex buildings

跨案例 · 跨形态

Across building cases
and architectural forms

03

降低模型
依赖度

Less reliance
on large models

工具与轻量模型协同

Geometry tools
& smaller models

04

轻量 BIM
交互式编辑

Interactive
lightweight BIM editing

查看 · 修改 · 反馈

Inspect · Edit
Feed back into design